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How to Start an XAS Experiment

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5.1 Introduction

As illustrated by this book, X-Ray Absorption Spectroscopy (XAS) has a very broad range of applications in different fields of science: physics, chemistry, material science, earth and environmental science, art and archaeology, biology, etc. Therefore, it is quite likely that scientists who hear about this technique for the first time (while looking in the literature or participating in a conference or just talking with some colleagues) might think about it as a useful instrument for their own research and would like to perform an experiment on their system. However, the first approach to XAS and to synchrotron light sources could leave an inexperienced user disoriented.

There are several books and reviews [1–10] (Chapters 2, 3, 11 of this book) on historical and more recent aspects of the XAS: theory, experiments, and data analysis will be supplied in the other chapters of this book. The aim of this chapter is to provide a short guide for beginners to help with the practical aspects that have to be faced previous to and during an XAS experiment in order to achieve the best results possible. The chapter is divided into three sections devoted to the main actions essential to the conduction of a successful XAS experiment: planning, preparation and performance. The experienced XAS users may already know most of the contents listed in this chapter but we hope that they find some new useful trick they did not know or they can learn new means to improve the outcome of their future synchrotron experiences.

For absolute beginners, you need to know that XAS uses a peculiar terminology to describe part of the spectra, or specific acronyms for variants of the technique so we will provide here a short vocabulary for you to understand the terms that will be used in this chapter:

Edge (E_0): point of the XAS spectrum where there is a sharp discontinuity in the absorption corresponding to the element-specific energy needed by an x-ray to extract a core electron. Commonly, it is arbitrarily

defined at the maximum of the first derivative or position at half-height of the jump. The nomenclature for edges is defined by the IUPAC notation based on progressive letters and numeral subscripts strictly correlated to the quantum numbers of the orbital from which the core electron is extracted (e.g. K-edge if a hole is created in 1s level; L₁, L₂ or L₃-edges if it is created in 2s, 2p_{1/2} or 2p_{3/2}, and so on).

Edge jump: the number defining the change in absorption in the XAS spectrum measuring the difference between the trends in the pre-edge and the post-edge regions at the edge position.

XANES (or NEXAFS): acronym for X-Ray Absorption Near Edge Structure (or Near Edge X-Ray Absorption Fine Structure, more common for soft x-rays). It is the part of the XAS spectrum developing in the range of few tens of eV around the edge. It may contain peculiar features such as: *pre-edge* peaks or shoulders, or a *white line* e.g. the peak with maximum absorption typically appearing just after the edge and some post-edge oscillations.

EXAFS: acronym for Extended X-Ray Absorption Fine Structure. This is the part of the XAS spectrum that extends up to 1keV or more after the edge. It contains several oscillations with the intensity decreasing while moving away from the edge.

First and higher shells: groups of scattering atoms divided according to their distance from the absorber. In order to describe the distribution of the atoms surrounding the absorber, in XAS, the space is described by concentric spheres of increasing radius. The atoms with shorter bond lengths belong to the first shell, while increasing the distance the other atoms are included in the second, third and higher shells. This division reflects the radial distribution visualized by plotting the FT of the EXAFS signal. So the atoms of the first shell will contribute to the first FT peak, and so on.

Beamline: the set of pipes, slits, optics components, instruments and walls that confine the x-ray beam to make it follow a line going from the magnets inside the storage ring where the photons are generated to the peripheral end-stations where the experiment is performed. A synchrotron usually contains several beamlines organized along a tangential direction with respect to the storage ring.

Monochromator: this consists of a couple of crystals (typically Si with a special cut along 111 planes) used to select the energy of the incident photons by means of Bragg's law. By changing the angle of the crystals, a different energy is selected.

Soft and hard x-rays: two sections of the wide range covered by x-rays in the electromagnetic spectrum defined to distinguish between the different behaviors of x-ray photons depending on their energy. Hard x-rays have higher energy sufficient to penetrate significantly through materials and to travel moderate distances through the air. On the contrary, soft x-rays have lower energies and are readily absorbed by most materials and air. A sharp distinction between the two ranges has not been univocally defined but the dividing line can be drawn approximately at 5keV.

5.2 Plan the Experiment

The planning of an experiment is probably the most underrated step of the whole process, probable reasons for this could be: the vast amount of preparation required to plan it with respect to the actual start of the experiment, the consequent difficulty in formulating clear ideas about how the situation will evolve, and the idea that any errors or inaccuracies at this stage will still be able to be amended along the way. Nevertheless, before starting any scientific experiment, it is of primary importance to have clear ideas about: (1) the objectives; (2) the information that can realistically be obtained; and (3) the procedures that will be followed for a successful conclusion. Moreover, the relevance of the planning is even greater in experiments to be performed on a synchrotron, where the most usual way to get time for measurements is through time-consuming peer-reviewed user programs.

5.2.1 Identify the scientific question

When studying a complex system, in whichever field of science, there are many unresolved issues. The questions can vary a lot on a case-by-case basis but typically they fall into one or more of the following categories:

- link macroscopic properties of a material to its atomic or electronic structure;
- understand the composition of an unknown material or find out if a particular specie is present or not;
- compare a series of similar samples where the change can be induced by: synthetic method, environment and external excitations (temperature, pressure, magnetic field, etc.) and time during a reaction.

In the following, you can find a list of scientific questions that are typically studied by means of XAS, divided into their research areas.

5.2.1.1 Physics

- *Semiconducting devices*: the study of the critical role of dopants measuring their local environment and understanding their location within the lattice (EXAFS) [11] (Chapter 16 of this book).
- *Superconductivity and strongly correlated electron systems*: highlight the consequences of small structural changes in the electronic properties (EXAFS), distinguish between the electronic states of chosen atomic species (XANES), study of the behavior at low temperatures (cryostat) [12].
- *Magnetic materials*: study of the spin and angular momentum order and how it is related to the magnetic properties probing different occupied (XES) and unoccupied (XANES) electronic levels, study of the behavior under a magnetic field (x-ray magnetic circular dichroism, polarized x-rays) [13] (Chapter 23 of this book).

5.2.1.2 Chemistry and Materials

- *Ion conducting oxides, batteries, and fuel cells*: monitor the changes in oxidation state (XANES) through all the stages of the devices' life and in real time, under operating conditions (QEXAFS) [14] (Chapter 18 of this book).
- *Catalysts and microporous solids*: study of the relationship between structure and functionality through relevant local structural information about the active sites (XANES and EXAFS). Determination of intermediates and products under operating conditions of high temperatures or under the flow of various gases (*operando*, QEXAFS or energy dispersive EXAFS) [15, 16] (Chapter 13 of this book).
- *Combinatorial chemistry*: comparison between a large number of compounds to test many formulations and optimize the chemical synthetic methods (XANES and EXAFS) [17].

5.2.1.3 Interfacial, Surface, and Nanoscale Science

- *Artificially-grown layered materials*: study of surfaces and interfaces *in situ* by probing a few atomic layers, determination of their properties by local structural information at the atomic level (grazing incidence and electron yield XAS); study of formation of organic thin films by directly measuring chemical bonding and energy band alignment at key interfaces (NEXAFS, soft x-rays) [18].
- *Molecules chemisorbed on surfaces and organic sensors*: study of covalent systems such as low-Z molecules, macromolecules and polymers, and their orientation on surfaces by investigating the direction of chemical bonds of the atom (NEXAFS, polarized x-rays) [19] (Chapter 26 of this book).
- *Nanomaterials*: study of the correlation between size and properties, measure of coordination number, interatomic distances and mean oxidation state of atoms for quantum dots in inert matrices or metal

clusters inside porous structures (XANES and EXAFS); study of the variations in chemical and structural composition at the nanoscale (nano-focus) [20] (Chapter 27 of this book).

5.2.1.4 Environmental and Earth Sciences

- *Soils and nuclear waste*: identification of the nature of toxic metals causing contamination in both solid and liquid environments, speciation with information about both crystalline and non-crystalline phases (XANES and EXAFS) [21] (Chapter 19 of this book).
- *Minerals*: speciation of all the phases in the mineral with particular interest in the structural development of amorphous or short-range order phases, and trace element substitutions (XANES and EXAFS), study of the local variations in oxidation state and elemental composition (micro- and nano-focus) [22] (Chapter 20 of this book).
- *Particulate matter (PM) and aerosols*: highlight the positive correlation between the local chemical properties of carbonaceous PMs and their sorption properties of toxic compounds (NEXAFS, soft x-rays), exploration of oxidation-induced changes in the viscosity and phase of liquid and solid atmospheric aerosols to improve the understanding of their chemical and physical aging processes (NEXAFS) [23].

5.2.1.5 Biological Sciences

- *Metal-based therapeutic agents and metalloproteins*: study of biomolecules in their natural environment such as physiological solution, structural information for partially ordered metalloproteins (EXAFS) and direct measure of oxidation state when it is in doubt (XANES).
- Study of the phases containing “soft matter elements” in biomolecular and pharmaceutical systems, organic and organometallic solution state chemistry and multicomponent products (NEXAFS, soft x-rays) [24].

5.2.1.6 Cultural Heritage

- *Archaeological artefacts*: study of efficiency of conservation techniques to investigate the changes in the oxidation state of objects of historical value (Chapter 21 of this book).
- Speciation of pigments, and unravelling details about how, when and where the artefacts were originally made (XANES and EXAFS) [25].

Nowadays, several techniques for the characterization of materials are available and it is important, before directing your attention to XAS, to ask yourself if there is any simpler way of obtaining the answers you need using more accessible instruments that frequently are found in any research laboratory. Thus, it is strongly suggested that you fully characterize your system by means of complementary approaches, such as:

- microscopy (SEM, TEM, STM, AFM, etc.) for magnification and internal visualization;
- other spectroscopies (UV-Vis, IR, Raman, fluorescence, NMR, EPR, etc.) or diffraction for a better understanding of atomic and electronic structure;
- analytical methods (spectrometry, gravimetric and volumetric analysis, etc.) to discover the composition of the sample
- theoretical calculations (*ab initio*, DFT, etc.) to generate models able to simulate the internal structure and properties of the material.

The data retrieved through these complementary techniques is important on its own, but it will play a key role also in helping the plan of XAS experiment and, at a later stage, in drawing the lines for a successful XAS data analysis.

5.2.2 Can XAS solve the problem?

First and foremost, in order to correctly determine the aims of the experiment and establish if it is the right tool to answer the question, it is important to know the potentialities of XAS and what kind of information can be extracted from the data.

XAS is an element-selective technique that consists of irradiating the sample with x-rays and measuring the absorption while scanning the energy of the incident beam along a range centered on a specific edge, which is dependent on the element and orbital object of the study. If an x-ray photon is absorbed by the sample, depending on its energy, it can promote the system to an excited state or extract an electron that propagates within the material and is scattered by the atoms near the absorber. Accordingly, the various regions of the spectrum (pre-edge, near edge or far from edge) contain information about the coordination geometry, oxidation state, and bond lengths of the absorbing atom. XANES provides more details about the electronic density of states and the coordination symmetry, while EXAFS enlightens about the radial distribution function of the surrounding atoms and their order/disorder. The information obtained is averaged over all the atoms of the same element inside the sample and if several constituents or atomic sites with different environment are present, all of them will contribute to the overall XAS signal. In this situation, the experiment is more difficult but it is still possible to identify the number of species and disentangle the single contributions by means of principal component analysis. Although it is possible to fit an EXAFS spectrum using specific codes to simulate the contribution of each component separately and then calculate the combined total signal, it is much easier to obtain this information by analyzing the XANES region where references can be used as fingerprints for the single species or site and the combination of the different contributions is linear.

As well as considering what XAS can do, it is essential to consider its limitations. Thankfully this technique is not destructive and it does not have strict limitations on the physical state of matter of the sample, which allows a wide range of applications. On the other hand, not all the systems can provide an adequate signal if investigated by XAS. The main constraint is the concentration of the element of interest, the lower limit is affected by many factors: the absorbing power of the main component of the sample (matrix or support), the available flux of the x-ray photon from the source, the sensitivity of the detectors used but it is rare that one can detect quantities lower than hundreds of ppm. If the atoms of the element of interest are contained inside a very absorbing matrix, only a very small fraction of the photons will be able to pass through the sample and reach the detector, or only a few superficial atomic layers of the sample will be probed, which could lead to the wrong interpretation of the system's physics. For example, if the sample contains 1000 ppm of Pd in a carbon matrix, the measurement is still feasible since the x-ray photons have enough energy to pass through the rest of the sample which will be almost invisible to them (the attenuation length of carbon at the Pd K-edge, 24 keV, is about 2 cm). On the contrary, if the sample contains 1000 ppm of Ti in CeO₂ it will be much trickier to get a good signal since the matrix is formed by high-Z atoms that will prevent x-rays with a low energy passing through it (the attenuation length of ceria at the Ti K-edge, 5 keV, is about 5 μ m, almost 4000 times lower).

The problems listed so far are mainly connected to the physical impossibility of measuring an experimental signal of good quality, but sometimes even if the spectrum looks perfect, it can be impossible to extract the information needed to solve the problem. The presence of many other atoms of the same element in the sample or in its support can make it difficult to obtain results regarding those sites which you are really interested in, since the interesting part of the signal will be overwhelmed by the other contributions. For example, if the sample is a semiconductor with two different binary alloys GaAs/AlAs separated by a ternary junction Al_xGa_{1-x}As, it will be impossible to obtain useful information on the As atoms at the heterojunction because the XAS signal at the As K-edge will be an average over all the As atoms in the sample, and the main contribution will come from the As atoms in the binary alloys which are predominant. Other limitations can be the lifetime of the photoelectrons, which restricts the range of distances the technique can probe to

5–6 Å, or the nature of the scattering atoms surrounding the absorber. Let us consider an experiment where the aim is to distinguish between two Cu-zeolites with same coordination around the Cu atom and different counterions (Na^+ and K^+) that stabilize the structures. A measurement at the Cu K-edge will not provide the information needed since the cations are usually situated in the pores of the material, at a distance higher than 5 Å from the absorber, so the probability of an electron scattering path Cu-Na or Cu-K is very low and consequently their EXAFS contribution is too weak. On the other hand, even if the difference you want to observe is in a short range around the absorber, if there are two atoms of the same element or of elements with a similar Z , at comparable distances, it would still be very difficult to distinguish between them since the scattering intensity is proportional to the atomic number Z . If the subject of the study is two linkage isomers $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]$ where the former has the NO_2 group bound to Co through nitrogen and the latter through one of the oxygens, EXAFS could theoretically note the difference in the first coordination shell. Nevertheless, in this case, N and O have similar atomic numbers 7 and 8, so they have a similar scattering power, moreover, there are other 5 N atoms in the Co surrounding at similar distances. Thus, the EXAFS signal generated by a Co coordination of 6 N or of 5 N + 1 O atoms will be comparable and, even with the finest experimental measurement, it will be almost impossible to distinguish between the two isomers. Other possible issues can be strictly related to the limits of the beamline where the experiment is to be conducted (energy range covered, intensity of the flux, spatial or temporal resolution, etc.) but they usually can be solved by applying for a dedicated beamline (see Section 5.2.3).

Bearing in mind the capabilities and limits of the technique just listed, it should be easy now to understand if in your specific case XAS is the right tool to solve your problem.

5.2.3 Select the best beamline and measurement mode

If you have read through the previous sections of the chapter and you still decide that a XAS experiment is the right approach to address your problem, then the next stage is to find the optimum place to perform it.

Synchrotron light sources are places where electrons are accelerated to approach the speed of light and then confined in a storage ring formed of straight and curved sections, where different magnets cause the generation of broad-spectrum x-rays. Synchrotron radiation is characterized by good intensity and the stability of the x-ray beam in a wide range of energies, qualities very important for a positive outcome in a XAS experiment. Thus synchrotrons are the most suitable light source for XAS. Over the years synchrotrons have developed [26] from a condition where the storage ring had generally particle physics purpose (first generation), to being designed for and devoted to only x-ray production (second generation). Then further improvements were made to enhance the intensity and the collimation of the x-rays flux by introducing insertion devices in the straight sections of the storage ring: arrays of magnets such as wigglers or undulators that act as additional sources of x-rays in addition to the usual bending magnets which are placed in the curved sections. Nowadays most active synchrotrons have these features built in or the synchrotrons have been upgraded to this level, also called the third generation. Looking to the future, a fourth generation of x-ray sources is available and can be divided into two categories: free electron lasers [27] (Chapter 22 of this book), characterized by extremely intense and short pulses that are very useful in reaching a sub-picoseconds temporal resolution but very damaging to the samples; and energy recover linear accelerators [28] that can function more like storage rings but with higher intensity. Figure 5.1(a) shows the huge improvement in photon flux over the years following the development of new generations of radiation sources.

There are more than 50 synchrotrons all over the world with different dimensions and energy imparted (see Table 5.1) [29]. The first step is to locate the most suitable light source in your area; the parameters to note are the level of innovation but especially the flux of photons it can provide at the energy range of interest for your study. Remember that synchrotrons operating at higher energies of the storage ring typically have a better flux in the hard x-rays range while smaller synchrotrons may be more suitable for soft x-rays measurements.

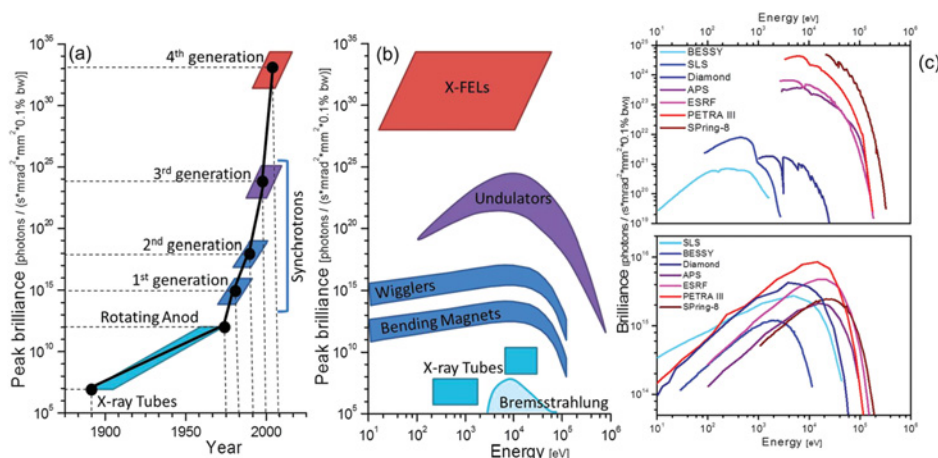


Figure 5.1 (a) Evolution of x-ray radiation sources and their brilliance along the years [30]; (b) Comparison between spectral brightness of Bremsstrahlung radiation, x-ray tubes, bending magnets, wigglers and undulators [31]; (c) Peak brilliance of some beamlines from the most intense x-ray radiation sources in the world [32].

When comparing x-ray sources, usually the intensity is measured in terms of brilliance (see Table 5.1) (Chapter 1 of this book), which defines the number of photons produced per second, considering also the angular divergence and the area of the beam and the fraction of photons included in a bandwidth (BW) centered at the energy of interest (Equation 5.1). A higher value of the brilliance will correspond to a better flux of x-ray photons on the sample; it is worth noting that the energy of the x-ray photons corresponding to the maximum brilliance is strictly correlated to the energy of the electrons in the ring (see Figure 5.1(c)).

$$\text{brilliance} = \frac{\text{photons}}{\text{s} \cdot \text{mrad}^2 \cdot \text{mm}^2 \cdot 0.1\% \text{BW}} \quad (5.1)$$

Once you have selected the synchrotron, then a better refinement is needed by pinpointing the beamline that is more suitable for the kind of experiment you have in mind. A beamline is generally formed of: (1) an optic hutch where slits and focusing mirrors are used to reduce the size and divergence of the beam, and a crystal monochromator selects the energy for the measurement through diffraction process according to Bragg's law; (2) an experimental hutch, where the sample is positioned and detectors or additional set-ups are in place to perform the measurement; and (3) a control room, where the user sits and controls the experiment using data acquisition software. The beamline configuration depends on the energy range covered. Hard x-ray beamlines usually have a path in air but shielding is needed to protect the user from radiation. Soft x-ray beamlines add a further difficulty to the experiment since the entire set-up (sample, detector, and diagnostics) must be contained in a vacuum up to ultra-high vacuum (UHV). The specs of the beamline depend on ring energy, the source magnet and the optics elements. Beamlines based on a bending magnet (BM) generally have a broader spectrum in energy, a lower intensity but high stability; conversely beamlines from insertion devices (ID) are characterized by a higher flux with a narrow bandwidth which results into a shorter energy range covered (see Figure 5.1(b)). When selecting the beamline, consider that ID beamlines are suggested for highly diluted systems since their highest flux could help to get a better signal to noise during the measurement. On the contrary, BM beamlines are more suitable for heterogeneous samples or if you need to measure several edges at different energies for their stability along a wide energy range. The optic elements, such as the crystals of the monochromator and consequently the Bragg angle adopted, influence not only the range of achievable energies but also the energy resolution of the spectrum which is higher if the d-spacing is smaller. If you are interested in studying very narrow features in the pre-edge region, look for a beamline that provides a

Table 5.1 *Some of the most intense synchrotron radiation sources operating all around the world and their specifications [33]*

Facility name	Location	Country	Energy (GeV)	Circum. (m)	Year
Australian Synchrotron	Melbourne	Australia	3.0	216	2006
LNLS	CNPEM, São Paulo	Brazil	1.4	93	1997
Canadian Light Source	University of Saskatchewan	Canada	2.9	174	2004
Shanghai Synchrotron Radiation Facility (SSRF)	Shanghai	China	3.5	432	2007
Beijing Synchrotron Radiation Facility (BSRF)	Institute of High Energy Physics, Beijing	China	2.5	240	1991
European Synchrotron Radiation Facility (ESRF)	Grenoble	France	6	844	1992
Soleil	Paris	France	3	354	2006
BESSY II	Helmholtz-Zentrum Berlin	Germany	1.7	240	1998
ANKA	KIT, Karlsruhe	Germany	2.5	110	2000
PETRA III	DESY	Germany	6.5	2304	2009
Indus 2	Raja Ramanna Centre, Indore	India	2.5	173	2005
ELETTRA	Trieste	Italy	2.4	260	1993
SPring-8	RIKEN	Japan	8.0	1436	1997
Photon Factory (PF)	Tsukuba	Japan	2.5	187	1998
Saga Light Source	Tosu, Saga	Japan	1.4	76	2003
Kurchatov Synchrotron Radiation Source (SIBIR1,2)	Kurchatov Institute, Moscow	Russia	2.5	124	1999
ALBA	Barcelona	Spain	3.0	270	2010
MAX-II	MAX-lab	Sweden	1.5	90	1997
Swiss Light Source	Paul Scherrer Institute	Switzerland	2.8	288	2001
National Synchrotron Radiation Research Center	Hsinchu Science Park	Taiwan	3.3	518	2008
Diamond Light Source	Oxfordshire	UK	3.0	562	2006
Advanced Photon Source (APS)	Argonne National Laboratory	USA	7.0	1104	1995
Advanced Light Source	Berkeley Laboratory	USA	1.9	197	1993
Stanford Synchrotron Radiation Lightsources	SLAC National Accelerator Laboratory	USA	3.0	234	1973
National Synchrotron Light Source	Brookhaven National Laboratory	USA	2.8	170	1982
Cornell High Energy Synchrotron Source (CHESS)	Cornell University	USA	5.5	768	1979

Table 5.2 Possible kind of XAS beamlines, measurement modes and most suitable samples

Beamline	Applications
Core XAS beamline	Good for XANES and EXAFS measurement on every kind of sample that doesn't require specific conditions and it is not too diluted. (ID beamlines recommended for lower concentrations). Typically used for measurements at the K-edge of transition metals or the L-edges of third row of transition metals and lanthanides/actinides.
Soft x-ray XAS beamline	Good for NEXAFS/EXAFS measurement at low-Z atoms K-edges or first-row transition metals' L-edges. Ideal for studying organic, biological or carbon-based materials or surfaces and interfaces.
Micro- and nano-focused XAS beamline (Chapter 10 of this book)	Ideal for micro- or nano-sized samples or for samples with a chemical composition varying at the micro- or nano-scale. Possibility of acquiring structural maps.
Quick-EXAFS (Chapter 7 of this book) or Energy dispersive EXAFS (Chapter 8 of this book) beamline	Ideal for operando experiments on catalysts or any other sample following in real-time reactions in the seconds or sub-seconds regime.
Time-resolved XAS beamline (Chapter 9 of this book)	Ideal for studying ultrafast processes, following in real-time electronic and geometrical transient of photoexcitable materials in a pump-probe scheme.
XAS and XES, high energy resolution beamline (Chapter 6 of this book)	Good for a full characterization of occupied and unoccupied electronic levels of the system. Possibility to acquire resonant XAS to enhance the energy resolution.

monochromator whose crystal cut has a smaller d-spacing (Si311: 1.6375 Å, Si400: 1.3578 Å, Si511: 1.0452 Å) as an alternative to the most common Si111 (3.1356 Å).

Each beamline is focused on more specific x-ray techniques, which can be absorption, fluorescence, diffraction, imaging, scattering, photoemission, etc. In the beamlines devoted to XAS, there are distinctions to be made considering the possibilities they offer to perform different experiments with various levels of complications: energy range, measurement modes and detectors available, particular spatial or temporal resolution, combination with other x-ray or spectroscopic techniques, or an additional set-up to study the sample under special conditions (external excitations such as magnetic fields, temperature, current or voltage, pressure, etc.). Table 5.2 lists some of the most common XAS beamlines and their optimal operation, measurement modes and most suitable samples.

If you are a beginner in XAS, probably you want to start with a simple experiment. The easiest and most standard experiment is performed on samples which are prepared in an external laboratory (typically pellets of the sample diluted with some blank powder) appositely for XAS measurement and then brought to the beamline and measured, as they are without any external excitation. This kind of experiment is called *ex situ*. Further levels of complication can be added and implemented in future experiments. The first step for a better understanding of the system is an *in-situ* study, to do it in its usual environment (e.g., inside a cell or mounted on a support or encapsulated in a matrix). The next step is to study the system while it is performing a reaction or under operating conditions, also called an *operando* experiment (e.g., a catalyst under the flow of the reactant gases at its usual working temperature). Practically every XAS beamline will allow you to perform a basic *ex-situ* experiment, but remember to check if the energy range covered is including the edge you want to study and if the detectors available are sensible at the levels of concentration of your samples.

A devoted experimental set-up may be used to improve the spatial and temporal resolution. For example, if the sample has differences in composition at the micro- or nanometer scale, you will look for a beamline which can provide a micro- or nanobeam, obtained by the insertion of special focusing mirrors, and then

acquire several XAS spectra in different regions of the samples. On the other hand, if you want to follow a reaction in real time, you will need a time resolution better than the time scale of the process of interest; in that case, you have several possibilities. Some beamlines now provide the possibility of performing quick EXAFS measurements [35,36] by continuously moving the monochromator to scan the energy and averaging the energy position and the photon counts during the acquisition time per point defined. (The most common alternative is a step scan where, on the contrary, the spectrum is collected point by point, stopping the monochromator for each data acquisition.) This method obtains measurement times down to 1, or even below one second or a few seconds to acquire a full XAS spectrum. A further improvement in the acquisition time of a XAS spectrum can be reached by energy dispersive EXAFS (EDE). The beamlines dedicated to this technique do not have a monochromator but use a bent crystal as a polychromator [37,38], which selects a range of energies and focuses the beam at the sample position, but then the photons of different energies diverge toward different points in a position-sensitive detector that acquires the full spectrum in one shot in milliseconds. This technique is widespread for *operando* experiments. To reach ultrafast time resolutions (pico or femto-seconds) [39], there are dedicated beamlines that adopt a chopper to select short x-ray pulses or that exploit the temporal structure of the beam resulting from the distribution of the electron bunches inside the storage ring.

Depending on how complex the information you are seeking is, a further enhancement can be obtained in energy resolution by adopting crystal analyzers to select the energy of the emitted x-rays and thus acquire resonant XAS spectra (also called High Energy Resolution Fluorescence Detection, or HERFD-XAS) [40]. This is possible in beamlines dedicated to the measurement of both x-ray absorption and emission spectroscopies.

Several variants of XAS have been developed over the years to obtain specific information about orientation (polarized XAS) [41], or magnetic (XMCD) [42], optical (OD-XAS) [43], and surface (grazing incidence) [44] properties of the sample, and so on. It is impossible to treat each of these and other very specific techniques in detail but there is extensive literature on them.

Your sample may need special conditions for measurement such as: high or low temperature, a controlled atmosphere, a magnetic field, applied voltage or current, high pressure, optical excitation, etc. Some beamlines make their equipment available to users: cryostats, furnaces, heating plates, temperature controllers, cells that allow gas stream through the sample, mass flow controllers, magnets, stirrers, voltage or current generators, diamond anvil cells, lasers, etc. And if you cannot find what you need in place or you prefer to use the same set-up you have in your lab, you can always take your instrumentation to the synchrotron, making sure you have the appropriate geometry and windows to mount it on the beamline without interfering with the x-ray path to the sample and then to the detectors. For the selection of the beamline, remember also to check if the beamline equips additional detectors or simultaneous probes. So if you think it would be useful to measure XAS and at the same time collect diffraction patterns or check the status of the sample with other spectroscopic techniques (IR, Raman, UV-Vis), or have an x-ray image of your sample or more, keep in mind that certain beamlines give you this facility.

Special attention should be paid to the different measurement modes available. The most common one and most performing in terms of quality of the signal and acquisition time for a spectrum is transmission. Practically every beamline is equipped with ion chambers as detectors for this acquisition mode. In fact, for transmission, the absorption is derived by measuring the transmitted I_T and incident flux I_0 , so at least two ion chambers are necessary: one in front and one behind the sample. Often a reference metal foil of the same element is placed behind the sample between the second and a third ion chamber (measuring flux I_{ref}), thus its absorption is recorded simultaneously. This precaution is useful for energy calibration and to check that the experimental conditions do not change along the life span of the whole experiment. Transmission is usually adopted for expressly prepared samples: powders in which you can control the amount pressed in a pellet, solutions where you can regulate the dilution or solids that can be cut with the right thickness so that you can optimize the total absorption. If the requirements for a good transmission measurement (*vide infra*) are not fulfilled, there are alternative acquisition modes such as fluorescence or electron yield that exploit

relaxation processes, which are secondary effects of the extraction of an electron from a core level due to the absorption of x-rays. The core-hole can be filled by other electrons in the outer electronic levels causing the emission of fluorescence photons with energies still characteristic of each element and electronic level, but lower than the incident photon's energy. These fluorescence photons are typically measured in Partial Fluorescence Yield (PFY) by means of a solid state energy-discriminating detector [45], which will be able to count only the photons in the energy range of interest and dismiss the other photons scattered or due to other element fluorescing. If you want to measure very thin samples or with a low concentration of the element of interest (doped glasses, catalysts embedded in matrices, trace elements in soils, highly diluted solutions), then fluorescence will be the most suitable method for you. On the other hand, electrons are also emitted: both photoelectrons ejected from core levels, and Auger electrons that, as part of the relaxation process as a higher-orbital electron fills the core hole, are analogous to fluorescence photons. It is possible to measure the Total Electron Yield (TEY) [46] collecting all the electrons by means of electron multipliers or by a grid close to the sample surface, or the Partial Electron Yield (PEY) [47], introducing an electron energy analyzer to select only the element-characteristic Auger electrons. EY may be an advantage for thin films or monolayers of a material grown on a substrate (since it will not suffer from background issues) or for light elements like C and N, bearing in mind that EY gradually decreases in favor of fluorescence as the atomic number increases up to $Z = 30$ where a cross-over occurs. This method works only for solid samples and, as the lifetime for electrons is much shorter than for photons, only the superficial part of the sample is probed.

Although it is difficult to choose the best measurement mode for your samples, being able to make the correct prediction is really helpful in selecting the right beamline, correctly preparing your samples and estimating the time needed to complete your experiment. To assess with a good approximation the XAS signal expected, several parameters have to be taken into account: the chemical composition, the density, and thickness of your sample. You will need to perform a calculation of the fraction of x-ray beam transmitted through the material I_t , with respect to its original intensity I_0 at two different energies: below and just above the edge. The corresponding XAS signal μx is derived by the logarithm of the transmission, according to the Beer-Lambert law (Equation 5.2) where μ is the absorption coefficient and x is the thickness of the sample:

$$\mu x = I_t = I_0 e^{-\mu x} \Rightarrow \mu x = -\ln \frac{I_t}{I_0} \quad (5.2)$$

You can then infer the maximum absorption and edge jump (difference between the μx signal above and below the edge), and compare the values found with the limits commonly assessed for a good outcome of the experiment, reported in Figure 5.2. This flow chart is meant to provide a rough guide that can help you in the selection of the measurement mode most suitable for your experiment. Remember that, if your conditions are close to the boundary defining one or another acquisition mode, often the possibility to collect at the same time in two different modes is available.

To perform the calculation of x-ray transmission through materials, you can use apposite websites [48]. In Table 5.3 some practical examples of absorption calculation are reported: for a Cu foil 8 μm thick, the calculated maximum absorption is 87%, lower than 95% and the edge is jump 1.55 which is higher than 0.1, so transmission is suggested (Figure 5.3(a)). For a 10mM water solution of copper nitrate $\text{Cu}(\text{NO}_3)_2$, contained in a 5mm thick liquid cell, the molarity corresponds to an atomic percentage of 0.018 at.% so we can use for the calculation the formula $\text{Cu}_{0.00018}(\text{NO}_3)_{0.00036}\text{H}_2\text{O}$, the maximum absorption is higher than 95%, so transmission is excluded. Considering that there is not a substrate, the best acquisition method is fluorescence; moreover, the calculated edge jump is lower than 0.1, thus fluorescence would be needed anyway due to the high dilution (Figure 5.3(c)). If the concentration of copper is higher, as for 0.5wt% CuO in cellulose pressed in a pellet 0.5 mm thick, corresponding to an atomic percentage of 1 at% $\text{Cu}_{0.01}\text{O}_{0.01}(\text{C}_6\text{H}_{10}\text{O}_5)$, the maximum absorption is lower than 95% but the edge jump expected is 0.07, thus is lower than 0.1 but close to the limit value. This is an intermediate case when it is difficult to state which is the best acquisition mode, the best solution would be to measure both transmission and fluorescence and evaluate the quality of the signal *a posteriori*. Fluorescence would be a safe choice because is certainly measurable, but if the homogeneity

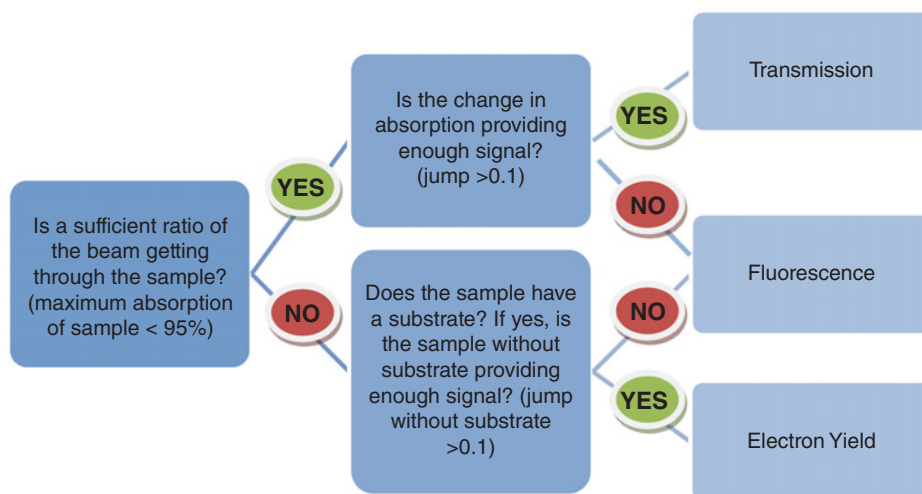


Figure 5.2 Flow chart to select the best XAS measurement mode.

Table 5.3 Examples of absorption calculation for three different Cu samples

	Cu Foil		CuO 0.5 wt% in cellulose		Cu(NO ₃) ₂ 10mM in water	
	Below	Above	Below	Above	Below	Above
Transmission	0.795	0.168	0.666	0.620	0.028	0.027
Absorption	0.205	0.832	0.334	0.380	0.972	0.973
μx	0.23	1.78	0.41	0.48	3.55	3.63
Max abs.	83.2%		38.0%		97.3%	
Edge Jump	1.55		0.07		0.08	
Thickness x	7 μm		0.5 mm		5 mm	

of the sample is good enough, then transmission can still result in a better signal (Figure 5.3(b)). For a more exhaustive explanation of absorption calculations, we refer to Section 5.3.2.

Besides selecting the detection mode, it is of strategic importance to choose the edge, the energy range, and the environment conditions in order to optimize the acquisition time and obtain the right information to address your specific scientific question. When you are trying to investigate properties related to the electronic levels of the system under study by means of XANES, remember that the electronic density that you will probe is restricted by three quantum selection rules that regulate the x-ray-allowed transitions: $\Delta n \neq 0$, $\Delta l = \pm 1$, $\Delta j = -1, 0, 1$. This means that if you want to focus mainly on d-orbitals, it is strongly suggested that you measure the L_{2-} or L_{3-} edges (hole in 2p) and not the K- or L_{1-} edge (hole in 1s or 2s) that will probe unoccupied p-orbitals. When you are interested in comparing the oxidation state of a specific element in a series of samples, then you may want to concentrate on the XANES region and collect shorter spectra, maybe with a higher energy resolution. If your aim is to reveal the coordination of an element to understand which or how many atoms are in the first shell, then you can measure a short EXAFS. If instead you want to disclose a complex structure, then you should acquire a longer EXAFS spectrum. Especially when looking at very fine details of the structure, further improvements can be obtained by increasing the acquisition time per point

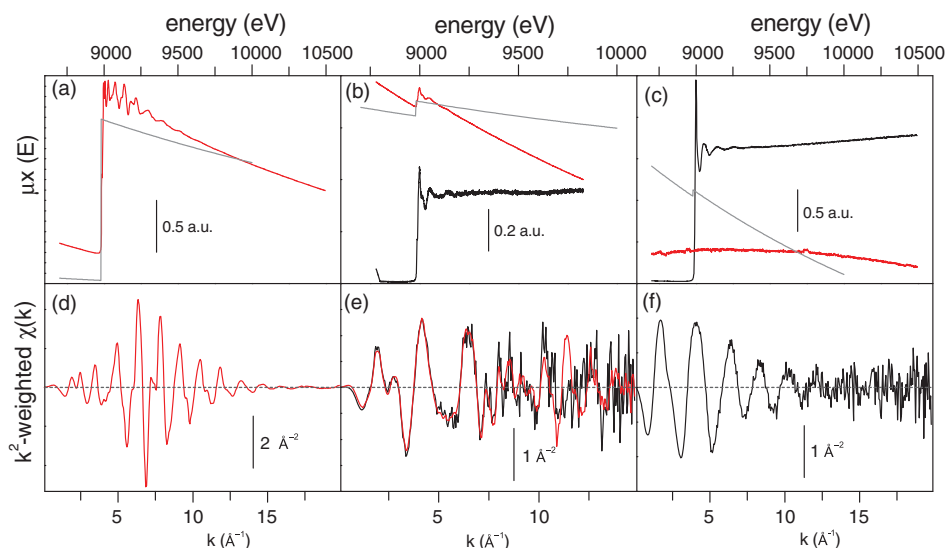


Figure 5.3 XAS spectra and EXAFS functions $\chi(k)$ for three different Cu samples. Dotted lines correspond to transmission measurements, black lines to fluorescence, dashed lines report the edge jump theoretically derived by transmission calculations starting from the composition and thickness of the sample. Parts a) and d) transmission XAS data for 8 μm thick Cu foil. Parts b) and e) fluorescence XAS data for 0.5wt% CuO in cellulose, pellet 0.5 mm thick. Parts c) and f) fluorescence XAS data for 10mM water solution of $\text{Cu}(\text{NO}_3)_2$ in a 5mm thick liquid cell.

to reduce the statistical noise as much as possible or by measuring at low temperature (e.g. in a cryostat) to reduce the thermal disorder which causes a broadening of the EXAFS features.

5.2.4 Writing the proposal

Once a critical evaluation has been made of the benefit of performing a XAS experiment, and after the rational selection of the best mode to perform it and of the source and instruments necessary, it is finally the moment to request the use of a beamline. In order to obtain time for an experiment, the last step is to write a proposal and submit an application. Usually two calls per year are issued to designate the beamtime for the next six months, so take into account that it may take a long time from the submission of the application to the conduction of the experiment. The proposals undergo a first assessment of the technical feasibility and safety, and then the scientific merit is evaluated by a peer review panel, composed of expert members external to the synchrotron. The beamtime will then be allocated according to the relevance given to the experiment by the panel and to the total operational beamtime available. It is worth noting that several synchrotrons offer funding to cover the travel and accommodation expenses of the experimenters.

Aware of the method of the beamtime assignment, it is obvious that, to increase the probability of obtaining beamtime, writing the proposal in a clear and detailed way is essential. First of all, the scientific question needs to be described briefly and simply so that even reviewers who are not specialists in the field can appreciate its relevance. Another point that has to be highlighted is why an XAS experiment is essential to solve the matter. Thereafter, you should describe how you plan to perform the experiment: which beamline you would need, which detector can be used, how you would mount the sample. Furthermore, specify the special equipment needed, if any special condition is required, ask for access to the laboratory for sample preparation or for the use of other instruments. It is a good habit to contact the beamline scientist in advance to discuss these details and to get suggestions about how to make the most of the beamline. It is recommended you provide

a list of samples and of the edges you want to measure for each sample and to realistically estimate the total amount of time needed, including beamline set-up, sample changes, and unexpected problems. Keep in mind that commonly at synchrotrons the time is measured in shifts where one shift corresponds to 8 hours. Finally, you should state your level of expertise and add a short description of how you will analyze your data, if you already have an idea of the approach to adopt.

An alternative route to obtain access to beamtime is through proprietary access, where you can pay for the time used and the services provided, depending on specific requirements. This route is commonly used by industrial users since it allows keeping the project and results confidential.

5.3 Preparing the Experiment

If your application has been successful and you have been assigned beamtime, then it is time to prepare yourself well in advance and define the last details for the experiment.

5.3.1 Experimental design

Going to a synchrotron for an experiment requires a vast effort in terms of physical and mental resources. Therefore, it is of the utmost importance to precisely select the team of people who will participate in the experiment. A good assortment of people with different competences and expertise would be very helpful to deal with the various tasks that have to be completed. Moreover, since the competition for beamtime is quite tough, usually the amount of hours allocated is as close as possible to the time essential to complete the experiment, therefore you will be forced to work day and night and try to make the most of the time at your disposal. Thus, the suggestion is to have a team formed of two to four people, depending on the overall duration of the experiment, including someone with a better knowledge of sample preparation and the scientific question and someone more expert in measurement and use of the software for data acquisition.

Other practical aspects have to be considered before moving to the synchrotron. There is a safety training procedure to perform in order to know the risks and have access to the light source, forms to obtain the authorization to use the labs, and booking for travel and accommodation. The synchrotron's user office will help you with the bureaucratic procedures, it will provide suggestions for the bookings and will send you reminders with adequate notice, so be prepared to be contacted by them.

Plan accurately the list of the samples and the order in which you want to measure them, try to group them by the characteristic of the samples but also by the edges with similar energies since it could take some time to set up the beamline for a different energy range. Try to give a priority to your samples and do not leave the most important ones to the end because accidents can happen and you may lose the opportunity to measure it. Keep in mind that XAS is a very good technique for comparison, so include in the list of your samples enough references containing the same element of interest with a well-known structure, possibly with different oxidation states, atomic ligands, or geometrical environment. They will be very useful both for XANES fingerprinting and linear combination analysis, and for disentangling EXAFS contributions.

Check with the beamline scientist if there are holders available for your samples. If you are planning to bring your own set-up, test it in your laboratories, check that you have spare parts in case something breaks during the experiment. Possibly visit the beamline before your beamtime to check if your set-up fits the available space for the sample environment and to perform a test with x-rays that will disclose if there is any problem with sample alignment or with the x-ray beam to cover the path to the detectors. If you have any issues either with the sample holders or with your experimental set-up, try to have a chat with the beamline technicians who can offer their experience in designing sample environments for x-rays, and can give suggestions on the best materials to use for shielding, windows, or frames.

Finally, always try to be flexible: be ready to change your schedule, prepare a back-up plan or bring some extra samples because unforeseen results or events could make you change your plans.

5.3.2 Best sample conditions for data acquisition

As previously stated, XAS is a valuable tool to study almost any kind of material: powders, solutions, gases, thin films, soils, glasses, sectioned solids, samples inside cells for in-situ environments, cultural artifacts, etc. Some of them may not need or allow any processing techniques or they can undergo only limited treatment, but more often than not they will require specific preparation to optimize their measurement, and it is a good habit to carry out these procedures in advance in order to save time during the experiment. The possible complications in the measure will reflect differently for each sample, depending on the way it has been prepared and on the acquisition mode. Most attention has to be given to the sample thickness, concentration, and uniformity that, if they are not ideal, may severely affect the data by introducing distortions in the signal, especially in the transmission mode [49]. Hereafter we will not only list the possible sample treatments and preparations but also we will try to explain the physics as the basis of the limits in the measurement. Thus, by exploiting a better knowledge of your system, you will have the tools to identify the major issues for your sample measurement and select the best actions to take in order to be successful.

We earlier hinted at the fact that transmission would be the ideal measurement mode; unfortunately it is also the one requiring the highest level of care for the preparation of an adequate sample. The physical quantities measured in transmission mode by XAS are the number of photons (or electrons) before and after the sample (I_0 and I_t), so the main limitations are having a sufficient number of photons transmitted and a reasonable difference across the edge. These quantities are regulated by the Beer-Lambert law (Equation 5.2) where the only parameter related to the sample that we can control is the factor μx . There are no units associated with this quantity but it can be expressed in number of absorption lengths which is a concept useful when comparing the thickness between samples with different concentrations, as it is probed by x-rays. One absorption length corresponds to the thickness that makes $\mu x = 1$ at the energy just above the edge of interest. Note that Equation 5.2 implies that the transmitted photons' intensity will be scaled exponentially with the number of absorption lengths n by a factor $(1/e)^n$; so one absorption length corresponds to the thickness that allows $(1/e) \sim 37\%$ of the photon to get through the sample, while the remaining 63% will be absorbed.

The absorption coefficient is a function of energy, and decreases when photons have higher energy since they have a higher probability of being transmitted through a material, with the exception of edges appearing because, when crossing a characteristic binding energy for a specific element, its absorption is much higher. So while the total absorption is proportional to the total amount of material, the edge jump, or $\Delta\mu x$, depends only on the number of atoms of the target element present in the sample (which, in turn, is defined by its thickness and concentration). The XAS fine structure, thus the signal that you want to measure, is proportional to the edge jump (Figure 5.4); if it is too low, the signal will be small, and you will get noisy spectra, but if it is too high, the transmission at energies above the edge will be much lower, increasing the error in the measure of I_t . A good value of edge jump to aim for is about 1 absorption length. The optimization of the edge jump plays a relevant role in maximizing the signal to noise, but the main limitation for transmission experiments is the total absorption of the sample, which is affected also by the nature of the environment, the matrix, or eventual support. If the sample is too thin, e.g., $\mu x \leq 0.01$, less than 1% of photons will be absorbed, resulting in a small edge jump and small signal as well, while I_0 and I_t counts and the noise derived by the propagation of uncertainties will be very high (Figure 5.4(c)). On the contrary, at the limit case of a sample with $\mu x \geq 5$, the transmitted photons will be less than 1% of the incident beam; apart from the increase in the statistical noise (the relative error is inversely proportional to the square root of the detector counts[50]), the main problem is that some unwanted photons may have a higher probability of reaching the detector, so their percentage which is generally negligible in the incident beam, may become relevant and introduce systematic errors in the measurement (Figure 5.4(d)). Examples of possible unwanted photons are: residual higher harmonics after the selection of the energy by Bragg's law; energy distribution tails of the not perfectly monochromatic beam (due to the intrinsic resolution of monochromator); or scattered photons that reach the detector without passing through the sample. Since there are problems if μx is too high or too low,

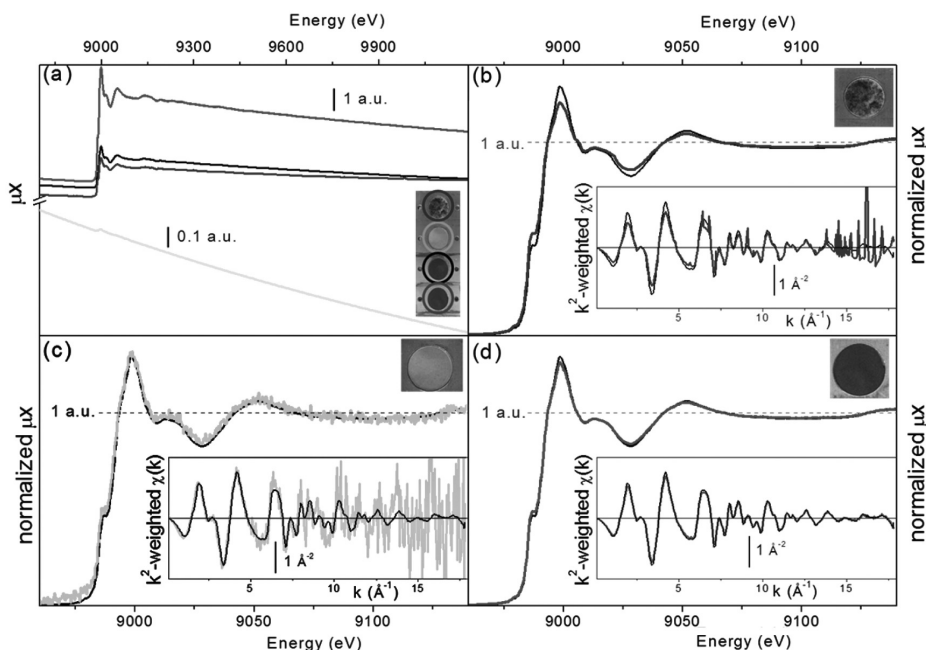


Figure 5.4 See colour plate section.

Table 5.4 Calculation of the mass absorption coefficient for CuO starting with the tabulated atomic absorption cross-sections σ_i

	Cu		O	Sample (CuO)	
	Below	Above		Below	Above
Weight (amu)		63.54	16.00		79.54
Weight percentage (w_i)		0.7988	0.2012		1
Cross section σ_i (g/cm ²)	37.88	288.97	7.88		
$\mu_m = w_i \sigma_i$ (g/cm ²)	30.25854	230.8292	0.7988	31.84	232.41

there is an ideal value for transmission measurements (or even better an accepted range, it is not crucial to get the precise value). The conditions recommended are a total absorption lower than three absorption lengths and edge jump between 0.1 and 1.5 [9, 10, 51, 53].

μx can be calculated according to Equation 5.3, knowing the density ρ , and can be optimized during the sample preparation by tuning either the thickness x or the amount of material used (mass m) and the concentration of the target element since the mass absorption coefficient μ_m is defined by the weight percentages for each element w_i and the atomic absorption cross-section σ_i that are tabulated [54].

$$\mu x = \mu_m \rho x = \mu_m \frac{m}{V} x = \mu_m \frac{m}{A} \quad \mu_m = \sum_i w_i \sigma_i \quad (5.3)$$

Hence let us try to calculate the optimal amount of CuO for the preparation of a pellet to be measured in transmission (see Table 5.4). You can use the support of specific websites[55–57] or software such as XAFSmass [58].

Once you have obtained the maximum mass absorption coefficient $\mu_m = 232.41 \text{ g/cm}^2$ and its jump across the edge $\Delta\mu_m = 200.57 \text{ g/cm}^2$, you can either optimize the thickness or the mass of the sample following Equation 5.4:

$$\left\{ \begin{array}{l} \mu_m \rho x_{opt} < 3 \\ \Delta\mu_m \rho x_{opt} = 1 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} x_{opt} < \frac{3}{\mu_m \rho} \\ x_{opt} = \frac{1}{\Delta\mu_m \rho} \end{array} \right. \text{ or } \left\{ \begin{array}{l} \mu_m \frac{m_{opt}}{A} < 3 \\ \Delta\mu_m \frac{m_{opt}}{A} = 1 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} m_{opt} < \frac{3A}{\mu_m} \\ m_{opt} = \frac{A}{\Delta\mu_m} \end{array} \right. \quad (5.4)$$

For CuO, whose density is 6.315 g/cm^3 , the resulting values suggest that if you have to prepare a thin film, the optimum thickness is $7.92 \text{ }\mu\text{m}$, and have a maximum allowed thickness of $20.5 \text{ }\mu\text{m}$. Repeating the calculations in the case of a pellet with an area (A) of 1.3 cm^2 , the optimal mass to be used is 6.6 mg and the maximum amount is 17.2 mg . During pellet preparation, the sample can be mixed with a blank powder composed of low-Z elements that will not have a big influence on the total absorption at the Cu K-edge. However, this was an easy example since we treated the case of a pure compound. In the case of diluted systems in a heavy matrix or films deposited on a thick support, you may not be able to use the optimal thickness or mass since it would be higher than the limit set by the maximum total absorption.

In these cases, the suggestion is to calculate the edge jump corresponding to the maximum thickness/mass allowed and check if it is higher than 0.1. If you cannot change thickness or mass in order to satisfy the optimal conditions, you can still use alternative acquisition modes. The decision should be taken case by case, since if the sample is very homogeneous, maybe even a smaller edge jump can still result in a reasonable signal.

If the sample is very diluted or thin, the fluorescence mode overcomes the limits of the transmission since it detects only the photons of interest that are generated after the absorption of the specific element so even if the signal is still small, the error on the measurement due to uncertainties on the counts and unwanted photons is reduced. Deriving a direct equation for the absorption coefficient when measured by fluorescence I_f (or electron yield I_e) is more complex and requires several steps and approximations. However, under thin and concentrated or thick and diluted sample conditions, the absorption coefficient is directly proportional to the ratio I_f/I_0 up to the addition of a slowly varying background. Although in these cases, fluorescence is the easiest and may be the only possible solution, there are drawbacks such as the higher acquisition time to reduce the statistical noise and obtain a reasonable signal-to-noise ratio. In fact, the number of photons counted may be much smaller with respect to transmission due to a limited solid angle covered with respect to the isotropic emission of fluorescence photons, furthermore, photons scattered or fluorescence from other elements could also reach the detector, increasing the noise. If the sample is too thick and concentrated, there are still some issues: in the case of a high number of photons, fluorescence detectors can saturate because they have a limit on how quickly they can count and the penetration depth of incoming x-rays is reduced, so that all of the photons over the edge risk being absorbed by a very low number of atoms that will be the only ones contributing to fluorescence, this introduces some non-linear effects known as *self-absorption* (Figure 5.5) [59]. Mathematical methods or different geometries can partially correct this effect but the best approach would be to avoid such a situation; one solution could be measuring in the electron yield mode which will probe only the first layers of the sample but without any linearity issues since Auger electrons escape the sample with higher difficulty. This characteristic will also reduce the interference from the substrate. The major issues with EY are: difficulty in measuring insulating materials since electrons will not be emitted easily; electrons have a shorter lifetime so the detector should be really close and a vacuum or a controlled atmosphere may be needed; only the surface of the sample will be probed so it has to be kept pristine.

Last but not least, especially for transmission, uniformity of the sample is an extremely important requirement for a good measurement. To evaluate if a sample is uniform or not, it is good to compare the dimensions of the grains with the absorption length; in a uniform sample, the grain size is considerably lower. If the

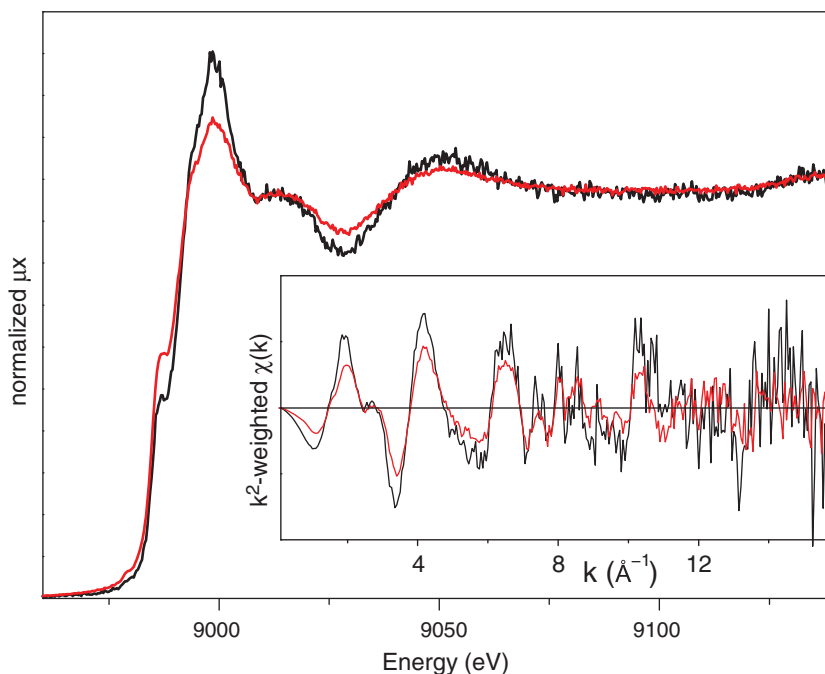


Figure 5.5 Self-absorption effect on XAS signal measured in fluorescence mode. Black lines correspond to a diluted sample, pale lines to a thick and concentrated sample. The intensities are suppressed both in the normalized μx and in the $\chi(k)$ EXAFS functions and the signal is distorted.

sample consists of granules larger than one absorption length, interpretation of the spectra is more complicated. Similarly, if the sample consists of several types of particles of different composition, the absorption length must be long enough that many particles of each type are sampled by the x-ray beam, otherwise errors of interpretation may result. In fact, if the area of the sample exposed to the beam has an uneven distribution of atoms, the signal measured would be the logarithm of the sum of all different contributions which is different by the average absorption coefficient (sum of the logarithms). Thickness effects always suppress EXAFS amplitudes, even when one normalizes the data by dividing by the measured edge step (see Figure 5.4). Normalizing the wiggles in this manner does tend to compensate somewhat for the spurious amplitude reduction, but it does not do the whole job. In addition, another problem that can introduce non-linear distortions in the XAS spectrum is the presence of pinholes or cracks in the pellet or foil that would allow a fraction of the photons going through the sample and altering the counts of transmitted photons on the detector (Figure 5.4(b)).

5.3.3 Sample preparation

Now that you have all the information, it should be easier to understand which is the best acquisition mode for your samples and which issues are more likely to be major, and so you can take the due precautions in the sample preparation phase. Hereafter you will find some examples of different materials and the suggested methods of optimizing the signal.

Cultural artifacts or rare materials may need to be kept intact, so they normally have to be measured in their genuine state in the fluorescence mode. If there is any region of interest or specific point in the sample

that you are more interested in, a good idea would be to create a mask that would allow it to be located more easily when aligning the sample on the beam.

Thin films are usually grown with some deposition method on a substrate, depending on the thickness with respect to the absorption length, the conductivity, and the interference from the substrate, they can be measured either with electron yield or fluorescence (if they are diluted, grazing incidence may increase the number of target atoms probed by the x-rays and reduce the substrate interference as well). If EY is selected, electropolishing or ion milling are the preferred methods to remove the surface layers that may be altered by the interaction with atmosphere.

Solution and gases are ideal for their intrinsic uniformity; they can usually be measured in transmission if a cell of the correct dimensions is prepared. The cell should have windows transparent to x-rays (thin kapton foils) and the material that it is made of should not interact with the sample. Good habits are: to test the endurance of the cell and if there is any leak, to have two holes in the cell for filling and for the free leakage of the air previously contained, to avoid the formation of bubbles; sealing the holes with tape or otherwise after filling may help to increase the endurance. You should also check if precipitate has formed in the solution and in which time range, if the time is longer than the length of a measure, shake the cell before starting, if it is shorter, think about a way to stir the solution while measuring. If you are interested in measuring a frozen sample, try to avoid the formation of crystals that could produce diffraction peaks interfering with the XAS signal.

Powders or particulate samples are probably the ones that require longer preparation, if the concentration and the nature of the matrix allow transmission measurements. There are several procedures to reduce the size of the grains and make the sample more uniform such as grinding, sieving, or by sedimentation. Grinding can be performed manually using a mortar and pestle in porcelain or agave or any material that ensures it will not chemically modify the sample, or mechanically by ball mills if you have a sufficient quantity of material. The simplest approach to separate particles of different sizes is to use a sieve, they are normally available with different mesh sizes but even with the finest ones, the best result will be to reduce the maximum size of the grains down to 20 microns. If a finer resolution is needed, than you can use sedimentation; mix the particles with an unreactive fluid and wait, the heavier particles will move toward the bottom of the container faster and after enough time you can pick out the smaller particles still at the surface with a pipette. After making the powder more homogeneous, you have to assemble the sample. One option is to press it into a pellet where the right amount of sample is mixed with a filler powder almost transparent to x-rays (boron nitride, cellulose, active carbon, sugar, etc.). You can perform calculations knowing the area of the pellet and the absorption coefficient to deduce the optimal mass of the sample to use, corresponding to the ideal thickness. The pellets can be enclosed in two pieces of tape to create a bag that is useful to keep the sample unchanged in the case of air-sensitive samples or just to avoid the powder falling onto the experimental table. Another method, more common if the amount of sample needed is smaller, is to spread the powder on a piece of tape by putting some powder on it and then uniformly spreading it with a spatula. Once an homogeneous state covers the tape, then remove the excess powder. If more thickness is needed, you can prepare several pieces of tape and superimpose one on top of the other.

For *in situ* or *operando* experiments, it is typically difficult to optimize the signal for transmission and the fluorescence mode is used more often. In these cases the problem is to find a compromise between the operating conditions of the material/device and the ideal amount of sample and conditions for data acquisition. If the experiment supposes gases passing through the sample, a powder as compact as possible would be ideal for getting the best XAS signal, since it is more uniform, but the presence of holes in the sample would be necessary for the gas to pass through it more easily. Additional environments can further complicate the measurement since they introduce other element of absorption or scattering of the photons, such as windows or a particular atmosphere. Moreover, they can induce changes in the sample, such as a higher thermal disorder if working at high temperature in a furnace, or it is the case of the crystallization of liquid solutions measured at low temperature, in a cryostat, with the consequent appearance of diffraction peaks, which would result in more noisy spectra.

5.4 Performing the Experiment

You should have learned by now that an XAS experiment requires a lot of preliminary work, starting months ahead. Once the dates fixed for your experiment approach, get ready to make the last effort in order to reap the benefits of this long preparation. Before leaving for the synchrotron, remember to put in your luggage or deliver by courier everything that you need for the experiment: samples, set-up or instrumentation (that you should have tested beforehand in your lab) and eventually spare parts or material in case something goes wrong. Once you arrive, you will be given instructions on the safety measures active on site and will obtain access to the beamline; before starting the data acquisition, take your time to discuss with the local contact and review your plans about the order of samples and measurement modes, so that the beamline can be set up and optimized for you.

5.4.1 Initial set-up and optimization of signal

For reliable data, the set-up of the beamline is fundamental. There are a lot of components and parameters to tune, related to optics and detectors, some of them are set by the beamline scientist at the beginning of the experiment and remain unvaried or they are modified only if there is a big change in the energy (e.g., to move to a different edge), while others need to be optimized for each sample. First of all, the x-ray beam has to be optimized at the edge of interest: the maximum flux of photons, and focus dimensions must be in the sample position. This tuning will be the responsibility of the local contact, who will be more familiar with the beamline operation, and this can take some time since it involves several steps: a scan of the monochromator, changing the angle between the two crystals to maximize the flux (the *rocking curve*), harmonic rejection through specific mirrors or slightly detuning the monochromator, and adjustment of mirrors and slits for the focusing. Consider suggesting a smaller focus, if you are interested in studying a particular point of the material or if you want to spot the differences in sampling various zones, while a defocused configuration is better if your sample has problems of uniformity, so that you can average out a bigger region. Also the detectors require preparation, so ion chambers need an appropriate gas mixture filling and voltage applied for the best response.

As soon as these tasks are accomplished, the energy can be calibrated by measuring a reference standard foil for the element of interest, and then you can finally mount your first sample. Considering that you want to perform some test on it, choose which sample will be the first with care. It would be desirable to have a copy of it or enough spare material to prepare the sample again in case it gets damaged, and to keep the possibility of measuring it again in optimal conditions. The selection will also depend on the kind of experiment you have planned but the suggestion is to try either a sample that has an average thickness and concentration or one of the trickiest ones: the most dilute, if you want to measure in fluorescence, the one with minimum edge jump expected, if you want to measure in transmission. In the same way, if your experiment requires the use of a special environment, try to do the test both in the simplest conditions and in the realistic operating mode to see if there are any problems introduced by the environment that are not ascribable to the beamline. You can first check if your material is subject to radiation damage by performing quick scans on the test sample and comparing the spectra obtained to see if there is any visible change. XANES is particularly sensitive and the acquisition is faster, you should look for shifts of the edge due to oxidation or reduction and for changes in the intensities of the features which give indications on the coordination geometry. If you note any sign of damage, think about the best way to avoid it: measure the sample in a controlled atmosphere (inert gas), mount a cryostat and measure at low temperature, or change the position of the beam on the sample at intervals shorter than the degradation time by moving the sample if it is solid or by flowing the sample with a pump, if it is liquid. Afterwards you can optimize the signal, a procedure that you may want to repeat every time you change the sample. Align the sample on the beam, rotate it to a favorable angle for your detector: perpendicular to the incoming x-ray beam if you measure transmission, at about 45° if you measure

fluorescence with the detector positioned at 90° . Then perform a scan of the stage where it is mounted, check that the beam is not clipping with the edges of the pipes of the beamline and of the sample holder or cell, and find the best spot where it is uniform and results in good counts. You can take your decision by plotting the counts of the detector as a function of the stage position. If you monitor the transmitted flux I_t , the intensities in the plot will be inverted with respect to sample thickness: you will see minima when the beam is more absorbed (e.g., when hitting the sample holder), and maxima when the fraction of beam transmitted is higher (if there is any empty space between the holder and the sample). If you plot fluorescence counts, the visual representation is straightforward since you have maxima when hitting a spot with more atoms of the target element and zero counts if you are in a hole, but you have to be careful since the holders may contain a small percentage of the same element in the sample. In both cases, try to select a point where the plot is rather horizontal or at least where the slope is not changing radically. Don't stay at the tip of a sharp maximum or minimum or you will see unexpected features in the spectrum if the position of the beam with respect to the sample changes even slightly during the measurement.

Finally, optimize the counts on the detectors: move the energy of incoming x-rays to where you would expect the maximum counts (just before the edge or at the very end of the scan for transmission) and set the gain of the amplifiers to ensure you get the maximum number of counts, but still at a due distance from the saturation limit to remain in the linear regime so that the output is proportional to the number of photons. Remember to check also the dark currents, measuring without x-rays, and take note of the numbers because it will give you an indication of the minimum counts acceptable for your samples. If you are using an energy-discriminating fluorescence detector, some additional steps are required. Move the energy above the edge, acquire a full fluorescence spectrum and define a window centered on the emission energy of the target element to read out only the counts that you are interested in. While optimizing the signal, try to move the detector closer to the sample to increase the in-window counts but remember to avoid hitting the saturation limit for the total counts. If the dominant counts in the fluorescence spectrum are given by scattered photons or by other elements fluorescing, you may consider using a filter in front of the detector. Typically to cut off the scattered photons, which give a peak at higher energy with respect to the fluorescent line of interest, you can adopt a low-pass filter containing the Z-1 element or any other with an absorption edge in between fluorescence and scattered energy. In this way, the photons you want to measure will be below the edge and will pass through, arriving at the detector, while the ones at higher energies will be absorbed more likely by the filter, being above its edge. On the contrary, to reduce the intensity of fluorescence produced by other elements in the material with lines at lower energies, you can use some aluminum foils or anything absorbing in the soft x-rays range. Since the absorption decreases exponentially as the energy increases, they will allow most of the photons of interest to pass through.

5.4.2 Data acquisition

It is now time to collect your real data, if you had clear ideas on what you want to disclose using XAS, you should tune the acquisition parameters according to the aims of your experiment. There are four main parameters for a spectrum and they are very simple: starting and final point, step size, and integration time. For the starting point, the ascertainment is simpler, it should be far enough from the edge to be able to extract and simulate a pre-edge trend, typically 200eV before the edge is a safe value. If you want to reduce acquisition times, you can risk and acquire a shorter pre-edge but it would be better to test it on one spectrum and try to do the data analysis to be sure that the range is long enough. For the final point, there are several aspects to consider, first of all, what data do you need? If you are interested only in oxidation state or coordination geometry that can be retrieved by XANES only, then the considerations to be done are similar to the ones already specified for the starting point, you should have a range long enough to be able to extract and simulate the trend of the post-edge, and then 300eV may be sufficient. Note that we want to simulate correctly the pre-edge and the post-edge trends in order to estimate as best as we can the edge jump and normalize all

the spectra. Correctly normalized XANES spectra are very useful because they easily allow a comparison of the different samples and then quantitative analysis. If you want to obtain good EXAFS results, you will need a longer range, that can go up to more than 1000 eV. However, the post edge is expressed more commonly not in energy but in k , the wave number of the photo-electron extracted, calculated by Equation 5.5 from the relative energy starting from the edge E_0 and having a dimension of 1/distance:

$$k = \sqrt{\frac{2m(E - E_0)}{\hbar^2}} \quad (5.5)$$

The k -range is a uniform scale to compare spectra acquired at different edges (and thus different energies). If the data needed is regarding only the first shell, a range up to 10–12 Å⁻¹ (400–500 eV) would be sufficient, if the interest is extended to a longer radius or for a better refinement of the interatomic distances, the range can ideally be extended up to 20 Å⁻¹ (1500 eV). Naturally, in some cases, the desire to obtain a lot of very precise information clashes with the possibility of obtaining a good signal until the end of the extended range. Considering that the intensity of the EXAFS oscillations decreases exponentially at higher k values, sometimes the noise overcomes the signal and it is useless to acquire in a long range, so the suggestion is to look at the data acquired and decide where is more convenient to stop. Take into account that the k -range, according to Equation 5.6, is also defining the number of independent points (N_{ind}) that is not equal to the number of measured data points (N). In most cases, the EXAFS spectra are significantly oversampled, so that $N_{ind} \ll N$. The limitation results from the fact that EXAFS spectra are “band-limited” and thus do not contain contributions from all possible frequencies:

$$N_{ind} = \frac{2\Delta k \Delta R}{\pi} \quad (5.6)$$

The number of independent points is important during the following data analysis because it limits the number of parameters that can be optimized by the fit of XAS data.

The definition of the step size instead limits the highest frequency that can be detected. By the Nyquist criterion of Fourier analysis [60], the data must be sampled at least twice in every oscillation of the highest frequency ($2 \delta k f_{max} = 2\pi$). Since the highest frequency is defined by the maximum scattering path length R_{max} that you want to measure, the sampling step δk should be at most π/R_{max} . If R_{max} is 10 Å, then the step should be smaller than 0.3 Å⁻¹. An oversampling by a factor 4–6 is suggested so that a typical step for EXAFS is 0.05 Å⁻¹, constant in k . Some beamlines work in quick EXAFS mode with a continuous movement of the monochromator and so the step size is constant in energy over the whole range and is mainly defined by the number of points, total acquisition time and speed of the monochromator. However, it is possible to select a small step and do a revision of the data *a posteriori* averaging out several adjacent points in the EXAFS region in order to obtain a proper statistical weight from the data. During a XAS acquisition, it is useful to apply a specific step size for different regions of the spectrum, changing also the acquisition time per point. The pre-edge region is not fundamental and is required only to extract the pre-edge trend, so long steps (5–10 eV) and short acquisition times (fractions of second) are sufficient. The XANES region typically needs a good energy resolution to detect also the tiniest features, the step here is defined considering the intrinsic broadening of the data due to the core hole lifetime. Typical broadenings range from 0.2 to more than 10 eV at increasing Z-numbers and are higher for K- with respect to L-edges. Even in the XANES, the suggestion is to oversample (0.1 to 0.5 eV are common values), always bearing in mind that the lower limit is also set by the monochromator angle resolution. The XANES signal is quite high, so also this region does not need a long acquisition time.

The EXAFS region instead is the most critical since, as previously stated, the signal quickly decreases so it is good practice to increase the acquisition time per point (1–10 s) while increasing the energy to reduce the noise. Another useful expedient is to repeat the measurement on the same sample at least two or three times to spot if there is any systematic error and to avoid the possibility that one of the spectra is corrupted

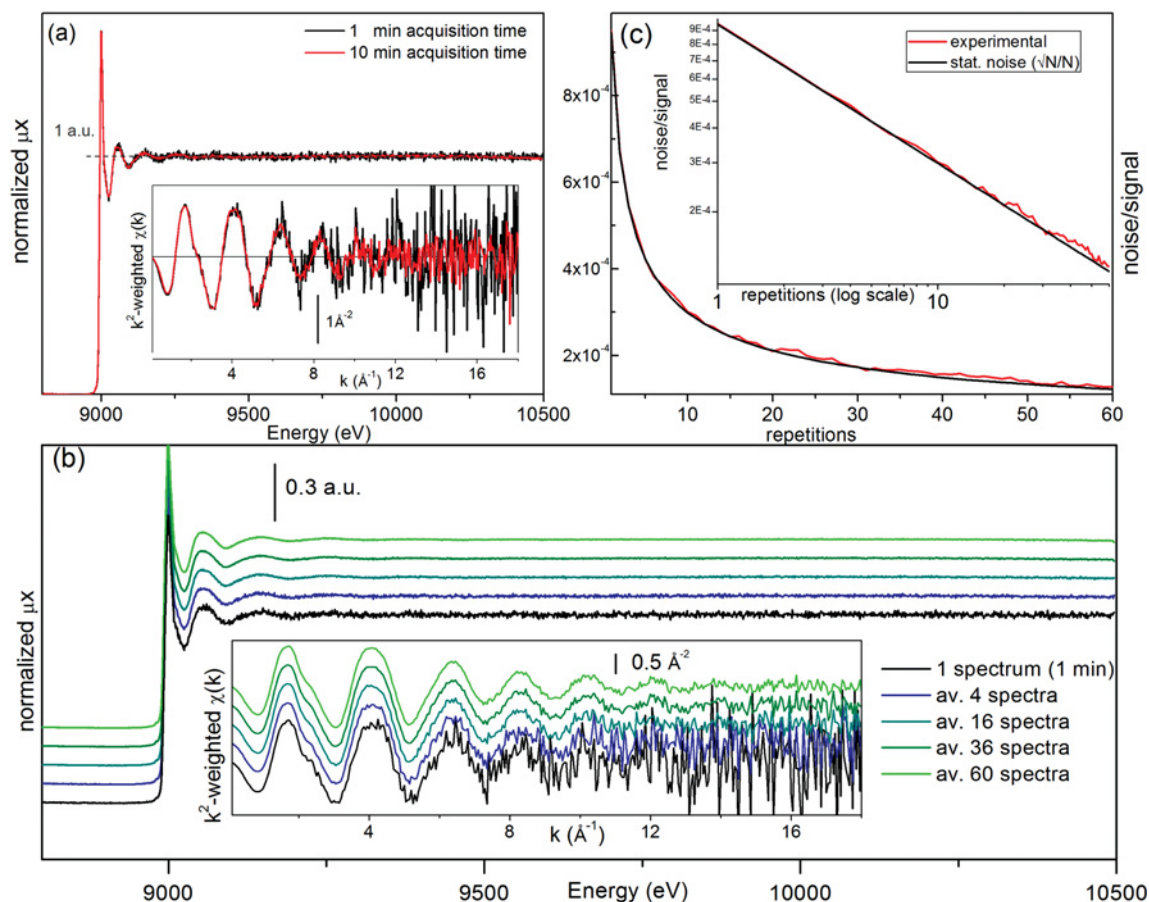


Figure 5.6 See colour plate section.

and you lose the data on that sample. Moreover, if you increase the total measurement time or if you average out the spectra collected on the same sample, you can reduce the statistical noise and obtain a better signal quality (Figure 5.6(a) and (b)). The number of repetitions can be tuned depending on the quality of the signal to noise that you need to achieve to extract the data needed on the sample. Keep in mind that the statistical noise is proportional to the square root of the counts, so after four repetitions you will have an improvement of a factor 2 and after 16 of a factor 4, thus, at a certain point, increasing the number of repetitions will not generate a relevant improvement in the quality of the signal and may cause the loss of important measurement time (Figure 5.6(c)). During the experiment you will certainly have some waiting time between two sample changes, the suggestion is to capitalize on this by performing a live analysis on the spectra just acquired. Software are available that give the possibility to plot either normalized XANES, EXAFS signal, and Fourier transform so you can get a first look at all the data contained in the spectra. In particular, by having a look at the EXAFS signal, you can evaluate the signal to noise and understand if a change in the final point of the acquisition range is needed. Cut at shorter energies if the noise in the last part of the spectrum is too high to save some precious time or, on the contrary, move it forward if there is still a signal that can be exploited. Moreover, if you average several repetitions on the same sample, you can understand how the statistical noise reduces and decide if you need more repetitions, and how much you can gain in the quality of the signal.

Normalized XANES and Fourier transform are instead useful to compare the samples and see if there are any evident changes in oxidation state or coordination, and in the radial distribution function. In this way you can have a quick glance and notice if there is some unexpected result that needs to be investigated more, or if you have a series of similar samples and you cannot note any differences; in both cases you may like to change the priorities of your experiment and reprogram the order of the samples to be measured.

In order to take advantage of all the time at your disposal, try to plan the duration of the measures according to the operation mode of the synchrotron; some of them operate in decay mode: the current in the storage ring is constantly decreasing and with it the flux of x-rays, and then a couple of times a day there is some dead time when the ring is refilled by injecting more electrons, other work in top-up mode with quick refills every few minutes. If possible, you would like to avoid having measurements going on when the refill is active, since it introduces a small instability in the beam size and changes the power load on the optics and counts on the detectors. While planning the time of measurements, you may like to consider the natural physical tiredness of the experimenters as well and try to leave long experiments running during the night, while, during the day, frequent sample changes are more affordable since the mental focus is higher.

In conclusion, once the experiment is finished, you will need to perform a deeper analysis and make the most of collected data, hoping that the suggestions given in this chapter may have helped you to invest your time and efforts in the best way to achieve the most useful results.

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